

Southeastern Section - 68th Annual Meeting - 2019

Paper No. 16-6

Presentation Time: 1:00 PM-5:00 PM

ACCURACY AND EFFICIENCY OF A GIS APPLICATION USING GOOGLE EARTH GEOLOCATION AND LIDAR DATA TO RECONSTRUCT AND SIMULATE ICE JAM FLOODING ON THE MOHAWK RIVER, NEW YORK

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The city of Schenectady, located in Upstate New York, experienced flooding due to an ice jam on the Mohawk River during the 14th and 15th of January 2018. The fluctuations in seasonal temperature during this time period created a blockage in the flow of the Mohawk River and thus the displacement of the water content. This research utilizes GIS software along with LiDAR and Google Earth elevation data, as well as publicly sourced imagery that depicts flooding locations and damages of the study area. We compared the accuracy of the flood reconstruction in Schenectady, NY utilizing two different elevation data sets; one from Google Earth and another from air-LiDAR data with spatial resolution up to 1 meter and 0.11 meters, respectively. Flood simulations were run in both LiDAR-DEMs and Google Earth 3D model to compare the flood-damaged areas and determine the corresponding water level during the flood event. The 3D model was flooded in Global Mapper several times within various regions of the study area in order to calculate the level of the flood at each neighborhood depicted in the available imagery. The standard deviation of the measured surface elevation from the ice jam flood images, using Google Earth and the Global Mapper simulation, is 0.21 meters and 0.25 meters, respectively. This research demonstrates that floods can be studied in a big data lab and it is not necessary to visit a damaged or flood-affected site. In addition, although LiDAR provides by far much better spatial resolution, the availability of orthorectified images makes Google Earth more efficient and accurate for decision making than Air-LiDAR. Therefore, we strongly suggest that a combination of high-resolution spatial data and imagery such as the capabilities of a drone mapping would provide higher accuracy than Air-LiDAR, and a more accurate flood simulation and evaluation can be developed.

Session No. 16--Booth# 6

[T12. Natural Hazards Prediction \(Posters\)](#)

Thursday, 28 March 2019: 1:00 PM-5:00 PM

[Carolina Ballroom \(Francis Marion Hotel\)](#)

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